

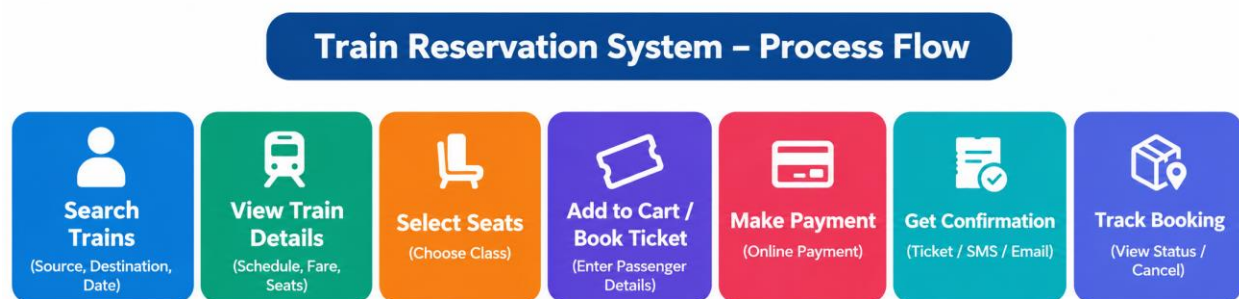
Week 1: Requirement Analysis and Customer Database Module

1. Requirement Analysis

Train Reservation System

A train reservation system manages the end-to-end process of searching, booking, cancelling, and managing train tickets. It allows passengers to search for available trains, check schedules and seat availability, reserve tickets, make payments, and track their booking status. The system also helps railway administrators manage train details, stations, routes, schedules, passengers, reservations, seats, and payments.

PROCESS CYCLE:



Train Reservation System – Process Flow

- **Search Train:** The Passenger searches for available trains by entering the source station, destination station, and journey date.
- **View Train Details:** After searching, the passenger can view train details such as train name, train number, departure time, arrival time, available seats, class, and ticket fare.

- **Select Seat / Class:** The passenger selects the required travel class and available seat according to their preference.
- **Book Ticket:** After selecting the train and seat, the passenger enters the required passenger details and confirms the reservation.
- **Checkout / Payment:** After confirming the booking, the system moves the passenger to the payment section. The passenger selects a payment method and completes the transaction.
- **Ticket Confirmation:** After successful payment, the system generates the railway ticket with details such as Passenger Name, Train Number, Journey Date, Seat Number, Coach Number, and PNR/Booking ID.
- **Track / Manage Booking:** The passenger can view booking status, check ticket details, download the ticket, or cancel the reservation when require

Entity	Description	Key Attributes	Key Relationships
Passenger	Person who searches and books train tickets.	passenger_id (PK), first_name, last_name, age, gender, email, phone	One-to-Many with Reservation
Train	Railway service available for passenger travel.	train_id (PK), train_number, train_name, source, destination, train_type	One-to-Many with Schedule, Reservation
Station	Railway station where trains arrive or depart.	station_id (PK), station_name, city, state	Many-to-Many with Train through Route
Schedule	Contains train journey timing and date information.	schedule_id (PK), train_id (FK), departure_time, arrival_time, journey_date	Many-to-One with Train
Reservation	Booking transaction created by a passenger.	reservation_id (PK), passenger_id (FK), train_id (FK), schedule_id (FK), booking_date, status	Many-to-One with Passenger and Train, One-to-One with Payment and Ticket
Ticket	Electronic ticket generated after successful reservation.	ticket_id (PK), reservation_id (FK), seat_number,	One-to-One with Reservation

		coach_number, class_type, fare	
Payment	Financial transaction associated with a reservation.	payment_id (PK), reservation_id (FK), payment_method, status, transaction_id, amount, payment_date	One-to-One with Reservation
Seat	Represents seats available in a train coach.	seat_id (PK), train_id (FK), coach_number, seat_number, class_type, seat_status	Many-to-One with Train
Admin	Person who manages the railway reservation system.	admin_id (PK), name, username, email, password	Manages Train, Station, Schedule, Reservation

Customer Table Creation:

```

SQL> CREATE TABLE Customer (
2     Customer_ID VARCHAR2(10) PRIMARY KEY,
3     Customer_Name VARCHAR2(50) NOT NULL,
4     Email VARCHAR2(50) NOT NULL,
5     Phone VARCHAR2(15),
6     Address VARCHAR2(100),
7     Registration_Date DATE
8 );

Table created.

SQL> INSERT INTO Customer
2 VALUES ('C001', 'Rahul', 'rahul@gmail.com', '9876543210',
3         'Chennai, Tamil Nadu', DATE '2026-09-10');

1 row created.

SQL> INSERT INTO Customer
2 VALUES ('C002', 'Ananya', 'ananya@gmail.com', '9876543211',
3         'Coimbatore, Tamil Nadu', DATE '2026-09-11');

1 row created.

SQL> INSERT INTO Customer
2 VALUES ('C003', 'Vikram', 'vikram@gmail.com', '9876543212',
3         'Madurai, Tamil Nadu', DATE '2026-09-12');

1 row created.

SQL> INSERT INTO Customer
2 VALUES ('C004', 'Meena', 'meena@gmail.com', '9876543213',
3         'Trichy, Tamil Nadu', DATE '2026-09-13');

1 row created.

SQL> INSERT INTO Customer
2 VALUES ('C005', 'Karthik', 'karthik@gmail.com', '9876543214',
3         'Salem, Tamil Nadu', DATE '2026-09-14');

1 row created.

```

Like this we can create as many rows we want.

```
SQL> SELECT * FROM Customer;
```

```
CUSTOMER_I CUSTOMER_NAME
```

```
EMAIL
```

```
PHONE
```

```
ADDRESS
```

```
REGISTRAT
```

```
C001      Rahul
```

```
rahul@gmail.com
```

```
9876543210
```

```
Chennai, Tamil Nadu
```

```
10-SEP-26
```

```
CUSTOMER_I CUSTOMER_NAME
```

```
EMAIL
```

```
PHONE
```

```
ADDRESS
```

```
REGISTRAT
```

```
C002      Ananya
```

```
ananya@gmail.com
```

```
9876543211
```

```
Coimbatore, Tamil Nadu
```

```
11-SEP-26
```

```
CUSTOMER_I CUSTOMER_NAME
```

```
EMAIL
```

```
PHONE
```

```
ADDRESS
```

```
REGISTRAT
```

```
C003      Vikram
```

```
vikram@gmail.com
```

```
9876543212
```

Update:

```
SQL> UPDATE Customer SET EMAIL = 'aakash35@gmail.com' WHERE CUSTOMER_ID = 'C001';
```

```
1 row updated.
```

Delete:

```
SQL> DELETE FROM Customer WHERE CUSTOMER_ID = 'C004';  
1 row deleted.
```